

DATA STRUCTURE LAB

LAB TASK 01



By

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Submitted to,

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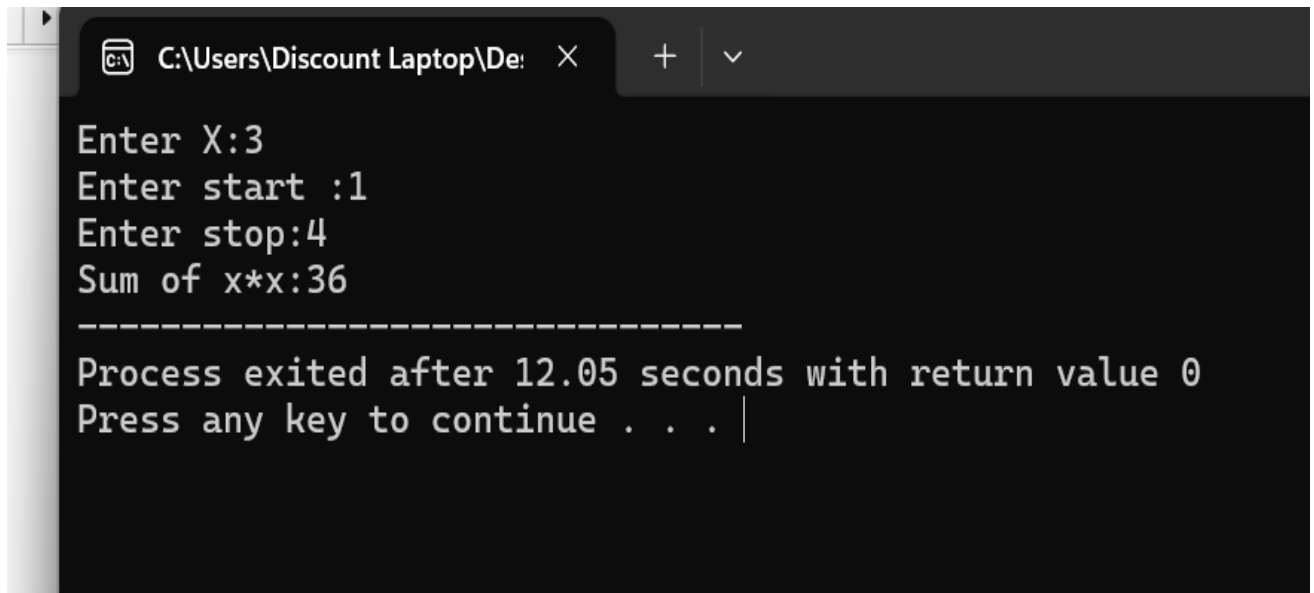
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LAB TASK 01

CODE:

```
#include<iostream>
using namespace std;
int main(){
    int start, stop,x,sum=0;
    cout<<"Enter X:";
    cin>>x;
    cout<<"Enter start :";
    cin>>start;
    cout<<"Enter stop:";
    cin>>stop;
    for(int i=start;i<=stop;i++){
        sum+=x*x;
    }
    cout<<"Sum of x*x:"<<sum;
    return 0;
}
```

OUTPUT:



```
C:\Users\Discount Laptop\De: X + v
Enter X:3
Enter start :1
Enter stop:4
Sum of x*x:36
-----
Process exited after 12.05 seconds with return value 0
Press any key to continue . . . |
```

LAB TASK : 02

CODE:

```
#include <iostream>
using namespace std;

int main()
{
    int a[50], n, x, value, i;

    cout << "Enter size: ";
    cin >> n;

    cout << "Enter values: ";
    for(i = 0; i < n; i++)
        cin >> a[i];

    // 1. Insert at end
    cout << "Enter value to insert at end: ";
    cin >> value;

    a[n] = value;
    n++;

    // 2. Insert at start
    cout << "Enter value to insert at start: ";
    cin >> value;

    for(i = n; i > 0; i--)
        a[i] = a[i - 1];

    a[0] = value;
    n++;

    // 3. Insert after specific value
    cout << "Enter value after which to insert: ";
    cin >> x;

    cout << "Enter new value: ";
```

```
cin >> value;
```

```
for(i = 0; i < n; i++)  
{  
    if(a[i] == x)  
    {  
        for(int j = n; j > i + 1; j--)  
            a[j] = a[j - 1];  
  
        a[i + 1] = value;  
        n++;  
        break;  
    }  
}
```

```
// 4. Insert before specific value
```

```
cout << "Enter value before which to insert: ";  
cin >> x;
```

```
cout << "Enter new value: ";  
cin >> value;
```

```
for(i = 0; i < n; i++)  
{  
    if(a[i] == x)  
    {  
        for(int j = n; j > i; j--)  
            a[j] = a[j - 1];  
  
        a[i] = value;  
        n++;  
        break;  
    }  
}
```

```
// 5. Display
```

```
cout << "Array List: ";
```

```
for(i = 0; i < n; i++)  
    cout << a[i] << " ";
```

```
cout << endl;
```

```
// 6. Delete from end  
n--;
```

```
// 7. Delete from start  
for(i = 0; i < n - 1; i++)  
    a[i] = a[i + 1];
```

```
n--;
```

```
// 8. Delete specific value  
cout << "Enter value to delete: ";  
cin >> x;
```

```
for(i = 0; i < n; i++)  
{  
    if(a[i] == x)  
    {  
        for(int j = i; j < n - 1; j++)  
            a[j] = a[j + 1];  
  
        n--;  
        break;  
    }  
}
```

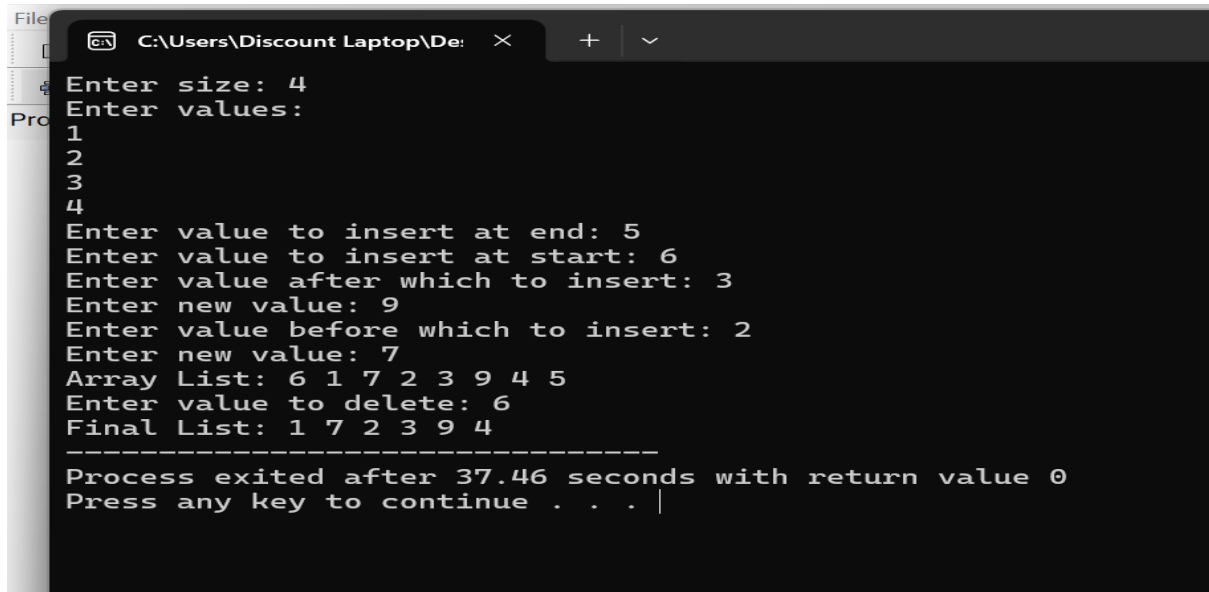
```
// Final List  
cout << "Final List: ";
```

```
for(i = 0; i < n; i++)  
    cout << a[i] << " ";
```

```
return 0;
```

```
}
```

OUTPUT:



```
File Edit View Help
C:\Users\Discount Laptop\De:
Enter size: 4
Enter values:
1
2
3
4
Enter value to insert at end: 5
Enter value to insert at start: 6
Enter value after which to insert: 3
Enter new value: 9
Enter value before which to insert: 2
Enter new value: 7
Array List: 6 1 7 2 3 9 4 5
Enter value to delete: 6
Final List: 1 7 2 3 9 4
-----
Process exited after 37.46 seconds with return value 0
Press any key to continue . . . |
```

LAB TASK 03

CODE:

```
#include<iostream>

using namespace std;

int main(){

    int data[10]={1,7,2,3,9,4};

    int size =6;

    int key;

    cout<<"Enter value to search :";

    cin>> key;

    int i=0;

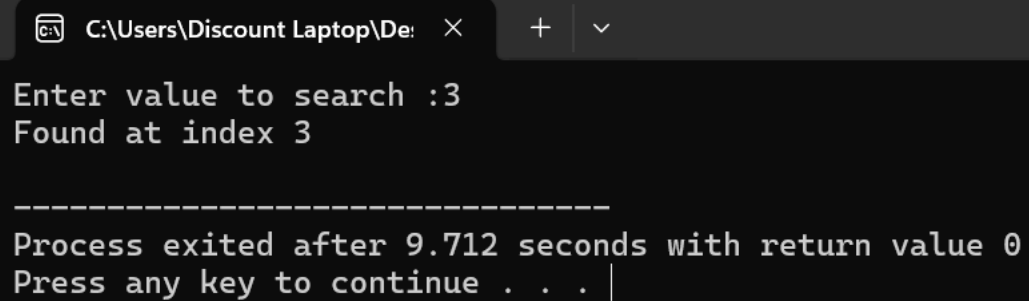
    while(i<size && data[i]!= key){

        i++;
```

```
}  
if(i<size){  
    cout<<"Found at index "<<i<<endl;  
  
}  
else{  
    cout<<"Not found"<<endl;  
  
}  
return 0;  
}
```

OUTPUT:

```
2 using namespace std;  
3 int main(){  
4     int data[10]={1,7,2,3,9,4};  
5     int size = 6;
```



```
C:\Users\Discount Laptop\De:  X  +  v  
Enter value to search :3  
Found at index 3  
  
-----  
Process exited after 9.712 seconds with return value 0  
Press any key to continue . . . |
```